IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

Corrected Copy

ROLE AND VALUE OF SMART GRID TECHNOLOGIES

Tuesday, 21 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions. All questions carry equal marks

Use answer book A to answer Question 1, answer book B to answer questions 2 and answer book C to answer question 3 & 4.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : , G. Strbac, F. Teng
Second Marker(s) : Teng, G. Strbac

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Question 1

In Szohôd, the capital of Borduria, a part of its 33 kV distribution network is facing challenges and the network operators and planners are considering alternative solutions. The 4-bus network used for this analysis is illustrated in Figure 1. All lines have a reactance of 0.1 p.u. Line 1-2 has a thermal capacity of 45 MW, while thermal capacity of other lines is 32.5 MW. Lossless DC load-flow analysis could be performed to identify the network problem and consider alternative solution options. The parameters of generators in the system are listed in Table 1 below.

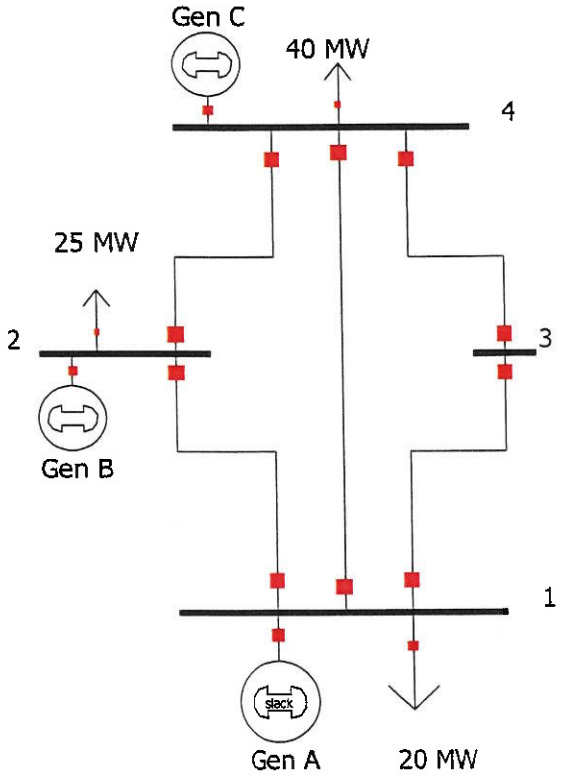


Figure 1: Szohôd distribution network

Table 1: Parameters of generators

Generator	Marginal cost (£/MWh)	Limit (MW)
Gen A	30	100
Gen B	50	20
Gen C	100	15

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Based on the network data, answer the following questions:

a) Determine the Economic Dispatch and compute the corresponding power flows in each of the circuits

[3]

b) Determine the power flows following the outage of each individual circuit and identify conditions in which the power flows during the contingency exceed the circuit capacity limit.

[3]

c) If the system operator needs to secure the system in a preventive mode against critical line outage (that causes the largest line overload) using the out-of-merit generator, determine the least-cost secured dispatch and the corresponding cost.

[6]

d) Another option is to install a Quadrature-booster (QB) in the system to reduce its operating cost. Determine the circuit in which the QB should be installed and the setting of the QB to ensure that security criterion is met for all single line outages (e.g. N-1 security standards are met).

[8]

Question 2

a) Explain the principles of centralised and price-based scheduling of flexible demand appliances. What are the advantages and disadvantages of each of these approaches?

[4]

b) A simple power system, operating at a two-hour horizon, includes:

- a generator producing power s_t (MW) at hour t with a cost function $C_t(s_t) = 100s_t^2$ (£) and a maximum output limit $s^{max} = 10\text{MW}$

- inflexible demand appliances, consuming power $D_1 = 3\text{MW}$ (at hour 1) and $D_2 = 2\text{MW}$ (at hour 2)

- 1000 identical flexible demand appliances with a shiftable cycle of duration $T^{dur} = 1$ hour, and demand $P^{cyc} = 5kW$. These appliances are activated at hour 1 and their cycle can be delayed by $\delta = 1$ hour.

i) Under centralised scheduling, 1) what is the optimal schedule of the flexible demand appliances at the two hours? 2) What is the total generation cost?

[2]

ii) Assuming that price-based scheduling is employed, 1) what should be the relation between the prices at hour 1 and 2 (denoted by λ_1 and λ_2 respectively)? 2) What is the resulting schedule of the flexible demand appliances at the two hours? 3) What is the total generation cost? 4) Is it higher or lower than the total generation cost calculated in i)? 5) Why it is higher or lower?

[4]

c) State the main big data challenges and the primary sources of big data in power systems.

[3]

d) Give three examples of big data analysis techniques. Give an example technique with an application in energy systems for supervised learning and unsupervised learning, respectively. For a transmission system operator, identify a specific task, describe how it was carried out in a traditional way and how it can be enhanced using big data.

[4]

e) Give a brief definition for concept of Demand Response (DR). What are the benefits/targets of DR? Give the definition of baseline estimation and illustrate the impacts of inaccurate baseline estimation. What are the current baseline estimation approaches?

[3]

Question 3

a) Discuss the concept of risk and its use in reliability analysis. Provide the definition of a risk metric used to quantify the reliability of distribution networks and one of transmission networks.

Explain why different metrics have been adopted to conduct reliability analysis for the distribution and transmission network.

[4]

b) Consider a section of a distribution network serving 12000 customers. The outage events occurred in one year are reported in the table below.

Customers Affected	**Duration**
3500	20 minutes
700	20 hours
8000	1 minute
1100	45 minutes
5000	3 minutes
10	1 day 3 hours
900	30 minutes

Table 2- Outage events

i) Compute CML and CI for this section of the network.

[2]

ii) Discuss the difference between CML and expected CML.

[1]

c) Consider the distribution network in Figure 2. The number of customers at each load point (a, b, c, d) is reported in Figure 2. The customers at load points a and b are of residential type. Each customer at load points a and b has an average power consumption of 0.4kW and a Value of Lost Load (i.e. interruption cost per unit of lost energy) equal to 5000£/MWh. The customers at load point c are commercial, while at load point d are of industrial type. Each customer at load point c has an average power consumption of 2kW and at load point d of 4kW. The Value of Lost Load at load point c is 20000£/MWh at load point d is 28000£/MWh.

The availabilities of the transformers (T1, T2) and the feeders (F1, F2, F3, F4) are reported in Figure 2. The availability of the Bus c-d is 99.95%. Assume that all components failures are independent and they all last more than three minutes. The individual capacity of T1 and T2 are always sufficient to serve all customers at load points a, b c and d. The network is radially operated but the normally open points (NOPs) can ensure some redundancy. They can be closed to resupply load points in case of faults in the network.

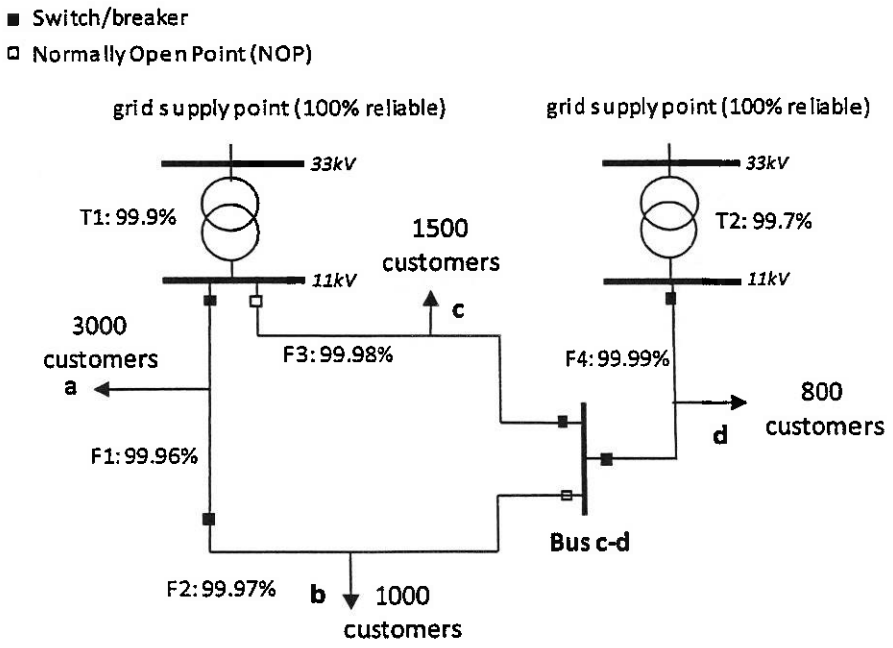


Figure 2- Network diagram

i) Calculate the probability that each load point is connected to a grid supply point (one expression for each load point).

[6]

ii) Compute the CML for the given network CML and the expected annual cost of interruptions for this network.

[4]

d) The Weibull distribution (Weibull 1951) is a generalization of the exponential distribution and is commonly used to represent component lifetimes in various types of engineering applications. The two-parameters probability density function is

$$f(t) = \lambda\beta\ (\lambda t)^{\beta-1}\ e^{-(\lambda t)^{\beta}}$$

where $\beta>0$ is a shape factor and $\lambda>0$ a scale factor

i) Compute the MTTF expression for a failure process modelled with the Weibull distribution. Hint: Remind that the gamma integral is defined as $\Gamma(k) = \int_0^{\infty} y^{k-1}e^{-y}dy$ and given k the value of the integral is tabulated. Given $\lambda = 0.01\ h^{-1}$ and the provided table draw the MTTF for a few values of β.

ii) Is the Weibull distribution memoryless?

(Hint: A distribution satisfies the memoryless property if $P\{T > t\} = P\{T > t + s \mid T > s\}$ for $t > 0, s > 0$, i.e the conditional probability does not depend on s.

iii) Compute analytically the hazard rate for the Weibull distribution and draw it for different values of the parameters.

(Hint: $\lambda(t)dt$ is the probability that a device of age t will fail in the small interval of time t to $t + dt$).

[3]

Question 4

a) There are two gas-fired power generating units. One (Generator 1) has failed 20 times in the past 8 years and repairs have taken 1 week on average, while another (Generator 2) has failed 10 times in the past 8 years and repairs have taken 2 weeks on average. Assuming that these are representative of their overall reliability, what is the mean time to failure (MTTF)? What is the availability of each unit? Compare and comment on the performance of the two units.

[4]

b) Consider a large-scale power station that has three operational states: 'failed', 'fully operational' and 'partially operational'. A Markov model is used to model transitions between states. The transition rates are as follows:

(1) from any operational state, complete failures occur at a rate of 1 per year;

(2) repairs from the failed state are carried out at a rate of 3 per month and always result in a fully operational unit;

(3) partially operational units return to full operation at a rate of 1 per month:

(4) fully operational units enter a partially operational state at a rate of 2 per month;

i) Draw the corresponding continuous-time Markov model.

[2]

ii) Draw a discrete time Markov chain for this process, assuming a time step of 1 day. Include all non-zero transition probabilities.

[2]

c) A system operator is designing a forced islanding scheme: in case of an acute blackout threat, the scheme preventively splits the power system into separate electrical islands. Such a scheme must have a high degree of dependability, but also be resilient against spurious activations.

The proposed design is as follows. The state of the power system is monitored at three locations, each equipped with a monitoring station. If a monitoring station detects a critical system condition it sends an alert to the control room over a dedicated communication channel. The decision to island the system is taken by a computer in the control room when it receives alert signals from at least two sites. To provide redundancy, there are three identical computer systems that receive identical signals and that can each activate the islanding signal.

Assuming that the critical condition is observable at each of the three monitoring stations (if they are operational), independence between components and an availability of 95% for monitoring stations, 98% for communication channels and 90% for the computer, quantify the probability that a critical operating condition is successfully detected.

[5]

A snapshot Monte Carlo simulation is used to estimate the expected cost of interruptions. After N=5,000 samples the best estimate of the cost is £108,762, with a standard error of £2581.

i) How many samples will be required (approximately, in total) to reduce the standard error of the mean to £1000?

[2]

ii) How many samples will be required (approximately) to reduce the width of the 95% confidence interval for the mean to £1000?

[2]

d) State examples of physical threats and cyber threats to reliable power system operation. For each category, provide one example of an accidental/natural threat and one example of a malicious threat (i.e. provide four examples in total).

[3]